Title :

Trader Performance vs Market Sentiment Analysis
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1. Objective

This analysis explores the relationship between Bitcoin market sentiment (Fear/Greed) and trader behavior on Hyperliquid. The goal is to identify patterns in trading performance and derive actionable strategies.

2. Methodology

- Loaded and cleaned both datasets
- Handled missing values and removed duplicates
- Converted timestamps and aligned datasets at daily level
- Merged trader data with sentiment data
- Created key metrics such as PnL, win rate, trade frequency, and trade size
- Performed analysis based on sentiment categories

3. Key Insights

- Traders exhibit higher risk-taking behavior during Greed periods, reflected in larger trade sizes and higher PnL variability
- Fear periods show higher volatility and inconsistent performance, indicating unstable market conditions
- Trade frequency increases during strong sentiment phases, showing reactive trading behavior
- Significant variation exists across traders, highlighting the importance of segmentation

4. Strategy Recommendations

- Reduce leverage and position size during Fear periods to minimize downside risk
- Avoid overtrading during Greed periods and maintain disciplined strategies
- Apply stricter risk management for high-frequency traders
- Use trader segmentation to implement customized strategies instead of a one-size-fits-all approach

5. Bonus

- Traders were grouped into clusters based on behavior and performance
- Identified high-risk and conservative trader segments
- This segmentation can be used to design targeted trading strategies